LEAD-19: the cofactor-projected factor base on the random-binary ladder

Date. 2026-10-23. **Status.** Measured under a protocol frozen before any run (`PROTOCOL.md`), two director-compiled task cards run by the mechanical executor and assessed `executed_as_specified`, every answer verified, the prime-order control counter-identical across arms, the raw arm reproducing the frozen round-5 counters exactly. **Result class (AGENTS.md §3). Engineering** on the relation phase of the random-binary rows of `docs/ic/LEADERBOARD.md` Table B: a different factor base, the same oracle, table, walk, linear algebra and verification, lowering the end-to-end cold `S` on the same instances by a measured, paired, bootstrap-bounded factor. An improvement to the baseline and so a success under the owner's criterion; not an advance, because every operation involved (group operations, `[h]`, a membership test, a hash table) is on the generic line and the family's `r^{1/6}` shape is unchanged. Every row stays above its matched rho, and the gap is reported as a distance. **Not yet a leaderboard row**: the validation gate's three artefacts are owed first (§8 below).

file	what it is
`PROTOCOL.md`	the frozen preregistration: arms, instances, commands, stop checks, predictions P1–P5, kill and success rules, the analysis plan and the §5a band
`cards/`	the two task cards, their receipts, the executors' verbatim reports and the director's assessments (`*.assessment.yaml`)
`char2.json`, `prime24.json`	card 1's raw records (five repeats; the control)
`char2_repeats20.json`	card 2's raw record (the pre-registered twenty-repeat extension)
`analysis.json`, `analysis.md`, `analysis_repeats20.*`	the frozen comparison's output for each record, by `docs/ic/tools/lead19_projected_base_compare.py`
`reports/run-20261023T2154Z/`, `reports/run-20261023T2201Z/`	the run reports (observations only) with graphs and PDFs
`plot_lead19.py`	the graphs' source (layout only)
`../pdp_literature_leads_20261022/leads/LEAD-19.md`	the dossier this protocol came from (desk estimates, prior art, obstructions)
`src/cryptanalysis/ic_boundary.rs`, `src/bin/ic/boundary.rs`	`--base-arms raw,projected,subgroup`: `project_factor_base`, `binary_subgroup_factor_base`, `prime_subgroup_factor_base`, `folded_table_class_census`, four unit tests

1. The question and the mechanism

The random-binary rows of Table B (`icv1-f2m15-t131-ed7256f0`, h = 2; `icv1-f2m21-t2591-de54d72e`, h = 2; `icv1-f2m27-t12543-569dca8b`, h = 6) are `mitm_m2_negfold_walk` rows: a negation-folded pair table of base-point sums, one walked target per trial, one hash probe. Their factor base is scanned from raw points of `E(F_{2^n})` with no projection into the prime-order subgroup `⟨G⟩`, while every walked target lies in `⟨G⟩`. A table entry `P_i + P_j` whose cofactor class `[r](P_i + P_j)` is not zero can therefore never match a target. The class census of the landed bases (measured by the run, P1) puts the useful fraction at 0.50, 0.50 and 0.17 on the three rows: half or five sixths of every table is dead weight.

Arm B replaces each abscissa's point `P` by `[h]P` (and its negative). `[h]` kills `E[h]` and is an automorphism of `⟨G⟩`, so the base, and every table entry, is in `⟨G⟩`; the relation yield per trial rises by about `h` at the same `|F|`, at the price of one scalar multiplication by `h` per abscissa (one doubling at h = 2; two doublings and one addition at h = 6) charged to the base phase, and no class test. Arm C keeps only points already in `⟨G⟩`, scanning a larger window to the same `|F|`; at h = 2 the membership test is the Kosters–Yeo trace parity `Tr(x) = Tr(a)`, at h = 6 or 8 it is `[r]P = 0`, charged.

The lever is measured against **A′**, the raw row minus the `[r]P` admissibility test the std rung carries (the test is a diagnostic the raw pipeline could skip; charging it to A would mix accounting into the lever). The ratio against the published rows is reported beside it as the accounting it is.

2. What ran

- Card 1 (`CARD-lead19-projected-base-20261023-1`): the frozen char2 ladder at degrees 15, 18, 21, 24, 27, five targets per curve (repeats 0 to 2 the frozen round-5 targets), arms raw / projected / subgroup at the std and balanced rungs, plus the 24-bit prime-order control; 36.9 s and 21.5 s.
- Card 2 (…`-2`): the same char2 command with `--repeats 20`, the extension PROTOCOL.md §9 pre-registered for a published-rung interval that straddles 1.10 (card 1's f2m15 read 1.00 with CI 0.82–1.21); 66.6 s.
- Stop checks (PROTOCOL.md §6), both cards: all rows verified; the prime-order control's projected and subgroup rows identical to the raw rows in `trials`, `relations_found`, `total_gae` and every phase counter; arm A's repeats 0 to 2 equal to the frozen round-5 counters on every char2 instance and rung.

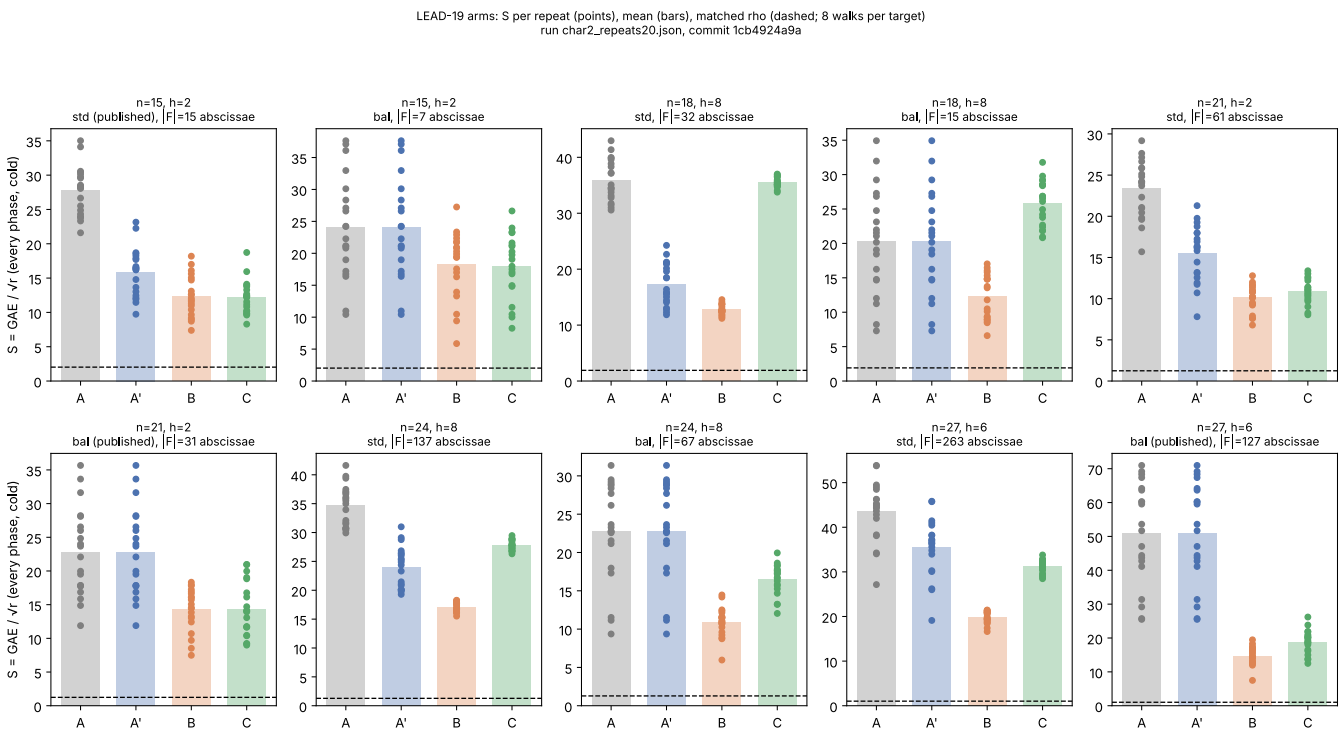
3. Results (card 2, twenty repeats; card 1 in brackets)

`S` = GAE/√r, every phase, cold, mean over repeats; ratios are per-repeat paired geometric means with the 95 % paired-bootstrap interval; "yield" is pooled relations per trial. The published rung is the leaderboard's.

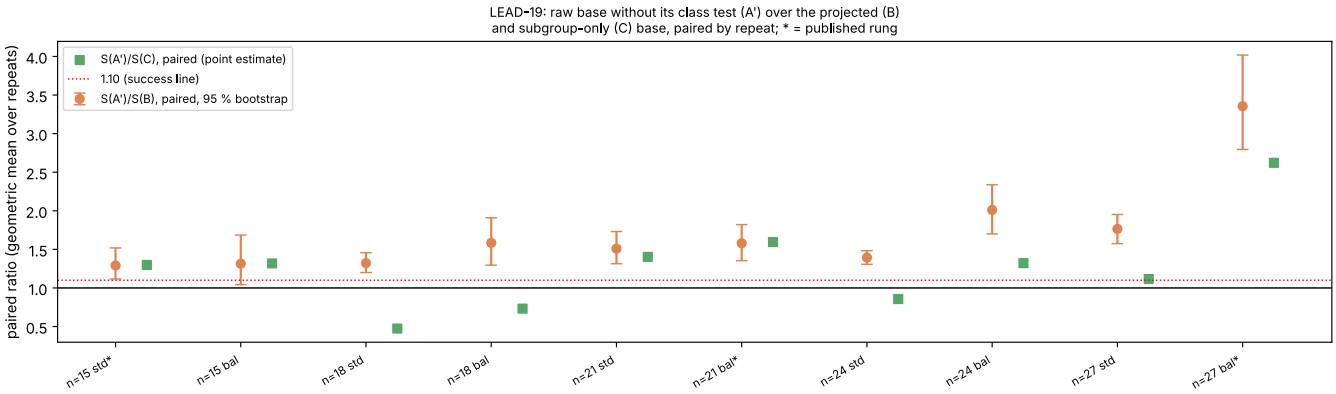
row	h	rung	S(A)	S(A′)	S(B)	S(C)	**S(A′)/S(B)**	S(A′)/S(C)	yield B/A	rho	B/rho (was)
f2m15	2	std (published)	27.75	15.88	12.35	12.16	**1.29 (1.12, 1.52)** [1.00 (0.82, 1.21)]	1.30	1.86	2.02	6.1× (13.1×)
f2m21	2	bal (published)	22.84	22.84	14.29	14.29	**1.58 (1.35, 1.82)** [1.54 (1.35, 1.77)]	1.59	1.96	1.24	11.5× (18×)
f2m27	6	bal (published)	50.99	50.99	14.75	18.84	**3.36 (2.80, 4.02)** [2.85 (1.73, 4.99)]	2.62	5.57	1.02	14.4× (38.2×)
f2m15	2	bal	24.02	24.02	18.20	17.98	1.31 (1.04, 1.69)	1.32	1.80	2.02	9.0×

f2m21	2	std	23.35	15.46	10.11	10.83	1.51 (1.31, 1.73)	1.40	2.03	1.24	8.1×
f2m27	6	std	43.57	35.54	19.80	31.26	1.77 (1.57, 1.95)	1.12	5.78	1.02	19.4×
f2m18	8	std	35.90	17.20	12.75	35.50	1.32 (1.20, 1.46)	0.47	3.78	1.91	6.7×
f2m18	8	bal	20.24	20.24	12.25	25.76	1.58 (1.30, 1.91)	0.73	3.67	1.91	6.4×
f2m24	8	std	34.66	24.02	17.08	27.79	1.39 (1.31, 1.48)	0.86	3.84	1.28	13.4×
f2m24	8	bal	22.75	22.75	10.94	16.49	2.01 (1.70, 2.34)	1.32	3.90	1.28	8.6×

- **Verdict of the frozen comparison** (PROTOCOL.md §9, read on the published rung): success on all three primary rows at twenty repeats (card 1 alone: n = 21 and n = 27, with n = 15 in the pre-registered extension). No kill condition fired: P2 (yield B/A within `[0.7h, 1.3h]`) holds on every primary row, the control did not move, and no S(A′)/S(B) ratio on a primary row reads below 1.10 (lowest published-rung interval bound 1.12).
- **Predictions.** P1 (class census 0.50 / 0.50 / 0.17) within 1 % (0.502, 0.501, 0.167); P2 1.86 / 1.96 / 5.57 inside the bands; P3 1.29 / 1.58 / 3.36 against the desk's 1.4–1.5 / 1.4–1.5 / about 3: f2m15 came in below the desk estimate (its base phase is half the row and B needs more relations to close, 219 against 194 over twenty repeats), the other two inside or above; P4 (B and C yields within 15 % at h = 2) 0.99 and 1.03; P5 exact.
- **Best rung.** B's best rung is the std base at f2m15 and f2m21 (S 12.35 and 10.11) and the balanced base at f2m27 (14.75); the paired best-rung ratios S(A′)/S(B) are 1.29 (1.12, 1.52), 1.51 (1.31, 1.73) and 2.41 (2.18, 2.67).
- **Pricing band** (AGENTS.md §5a). The only unpriced native work in the comparison is arm C's trace parity; at 0, 0.1 and 1 GAE per test S(C) moves by at most 0.25 and every A′ > B and A′ > C order holds, so the ranking stands across the band. Rho carries no unpriced work.
- **Yield.** On the primary rows every arm realises its exact counting ceiling (`yield_over_ceiling_exact` 0.98–1.08 at the published rungs): the projection raises the ceiling (all pairs class-zero) rather than the realisation, which is the mechanism the dossier predicted.
- **B against C.** Equal at h = 2 (same yield; C's selection is a free parity and a scan about twice as long). B wins at h = 6 and 8, where C's `[r]P` membership test is a charged scalar multiplication per candidate (1,547 of them at f2m27 std): at the std rung C's base phase is `S` 29.2, 24.1 and 26.5 on f2m18, f2m24 and f2m27 against B's 6.4, 13.3 and 14.9 (1.8 to 4.6×), while the relation phases of B and C match (6.2 and 6.2, 3.7 and 3.7, 4.9 and 4.7). B is the recipe.



S per arm, rung and repeat for the twenty-repeat run (points: repeats; bars: means; dashed: matched rho)



Paired ratio S(A′)/S(B) with its 95 % bootstrap interval and S(A′)/S(C), twenty repeats; red dotted line: the 1.10 success line

Figure sources: `plot_lead19.py` reading `analysis_repeats20.json`; card 1's five-repeat graphs are under `reports/run-20261023T2154Z/`.

3.1 What is lever and what is accounting

row (published rung)	frozen A (round 5)	B now	published-row ratio	lever S(A′)/S(B)	accounting
f2m15 std	26.31	12.35	2.13	1.29	the std rung's class test (1,516 GAE per target, 12 of A's 27.75 units) is not charged to B: that part is accounting, A′ carries the same scan
f2m21 bal	24.39	14.29	1.71	1.58	the balanced rung carries no class test; the difference between 1.71 and 1.58 is the frozen row's three-seed mean (24.39) against this run's twenty-repeat A (22.84)
f2m27 bal	36.93	14.75	2.50	3.36	no class test; this run's twenty-repeat A (50.99) is heavier-tailed than the frozen three seeds (25.5, 59.5, 25.7)

Only the lever column is claimed. The cost B adds is in the record: `projection_doubles` 15 / 61 / 526 and `projection_adds` 0 / 0 / 263 on the std rungs (one `[h]` per abscissa), against relation phases of thousands to hundreds of thousands of GAE.

3.2 Distance to rho

row	published rung	B	matched rho	gap now	gap on the leaderboard
f2m15	std	12.35	2.02	6.1×	13.1×
f2m21	bal (best: std 10.11)	14.29	1.24	11.5× (8.1× at std)	18×
f2m27	bal	14.75	1.02	14.4×	38.2×

The gap is the distance remaining, not a gate (AGENTS.md "Success criterion"). The family's shape stays `r^{1/6}` in the exponent (ledger §11.2); this lever is a constant.

4. Controls and checks

- **Negative control** (`icv1-fp24-t1577-77336155`, h = 1): arms B and C are the identity by construction and the run shows them counter-identical to A in every phase (S 24.9 std, 14.3 balanced, all three arms).
- **Frozen counters**: arm A's repeats 0 to 2 reproduce `docs/ic/runs/ic-boundary-ledger-round5-2026-09-22.json` exactly (`trials`, `relations_found`, `total_gae`) on every char2 instance and rung, so the raw rows of this run are the ledger's rows and the pairing is a comparison, not two runs.
- **Default path unchanged**: `--base-arms raw` leaves the record as it was (the configuration key is omitted; the unit test `lead19_char2_arms_run_and_verify` checks no arm row and no census appear).
- **Verification**: every row recovers the planted logarithm; the control's and every char2 row's `verified` is true in both records.

5. Literature and prior art checked

- Kosters, Yeo, *Notes on summation polynomials*, arXiv:1503.08001v3 (2015), Prop. 4.2 and Remark 4.8: the trace map `P ↦ Tr(x(P) + a)` has kernel `2E`, and relations may be sought in that kernel. Read at source for the dossier; the trace test of arm C is that remark made an arm.
- Fong, Hankerson, López, Menezes, *Field inversion and point halving revisited*, IEEE Trans. Computers 53(8) (2004), Lemma 3.1.4: `(x, y) ∈ ⟨G⟩ ⇒ Tr(x) = Tr(a)` for `G` of odd order. The unit test `lead19_projected_and_subgroup_bases_lie_in_the_subgroup` checks the equivalence on every lifted abscissa of a 64-element window at h = 2.
- This repository: `koblitz_index_calculus.rs` `build_subgroup_orbit_factor_base_with_cost` already multiplies by the cofactor on Koblitz bases ("so no sample is wasted"); the random-binary ladder had no such hook (`BinarySubspaceBase`, `binary_subspace_factor_base`). `research/isogeny_conductor_gap_ic_20261010` measured the fiber-aware oracles (lifts, closed tables) on cofactor curves: yield up by about `h` but `S` not, because they pay `h` probes or an `h×` table per target; B gets the yield with the same table and one probe. KN-FIND-007 (yield conservation, `C(B+m−1, m)/N`) fixes `N` as the order of the group the sums live in; moving the base into `⟨G⟩` changes `N` from `#E` to `r`.

- Dossier: `../pdp_literature_leads_20261022/leads/LEAD-19.md` (desk estimates 1.4–1.5× at h = 2, about 3× at h = 6; obstructions O1–O8).

6. Honest limits

- The matched rho is the ledger's own negation walk in the same process, eight walks per target; its mean has its own spread, and the gap figures move with it.
- The raw arm's `S` is heavy-tailed (f2m27 balanced: 25.5 to 71.0 over twenty repeats); the paired ratio absorbs the target-to-target part of that, the bootstrap interval the rest.
- f2m15's lever (1.29) is below the desk's 1.4–1.5: at `|F| = 15` the base and table are half the row, and B's relation graph closes later (the dossier's O4). The card-1 interval straddled 1.10 and only the pre-registered extension decided it.
- The secondary rows (h = 8) and the m = 3 rows were not pre-registered predictions; their numbers are reported, not claimed.
- Nothing here is a replay certificate, a validator report or a `VALIDATION.json`; the leaderboard rows keep their round-5 figures until those exist (§8).

7. Classification

Engineering (AGENTS.md §3: `S` fell, the method stays on the generic line) on the relation phase of the random-binary rows: the base is projected into the subgroup, the pipeline's end-to-end cold `S` fell on every primary row with every answer verified, under a protocol frozen before timing and a kill rule that did not fire. It is an improvement to the baseline and so a success under the owner's criterion; it is reported as engineering and not as an advance, since the oracle is a pair table and a hash probe and nothing here moves the family's `r^{1/6}` shape. The published-row ratios above 1.29 / 1.58 / 3.36 are partly accounting (the std rung's class test) and partly the frozen rows' seed means, and are not claimed.

8. Next steps, in the order they decide the most per unit of cost

1. **Land the rows through the validation gate** (`docs/VALIDATION_GATE.md`): a replay certificate (an independent re-run of the two cards on another host; the run is deterministic in the seed, so the counters must repeat bit for bit), a validator report recomputing `S`, the paired ratios and the band from the raw records, and a `VALIDATION.json` naming the matched negation rho and the §5a band; then regenerate the leaderboard with B as the Table B recipe (f2m15 12.35 std, f2m21 10.11 std, f2m27 14.75 balanced) and a `benchmarks/` entry.
2. **Make `projected` the ladder's default for cofactor curves** once the rows are validated, and re-run the balanced-rung search with B (the dossier's desk model put B's best rung at the std base on f2m21; the run agrees there and at f2m15, and finds the balanced base best at f2m27).
3. **m = 3 rows.** The same projection makes every three-summand probe `h×` more productive; not a leaderboard recipe today, cheap to measure with the arms already in place.
4. **Koblitz Table A/C rows**: confirm which recipe each used (`compact-orbit-scan` and `build_subgroup_orbit_factor_base` already project; the dossier did not open every row).